Introduction to Linux OS

Open-Source | Community Developed | Multiple Users

What is LINUX?

Linux is a Unix-like, open source and community-developed operating system for which is capable of handling activities from multiple users at the same time.

Easy to Use

Portable

Multiple Operating Environments

What is Windows?

Windows is a licensed operating system in which is designed for the individuals with the perspective of having no computer programming knowledge and for business and other commercial users.

01 RED HAT ENTERPRISE LINUX

redhat

Commercial Linux distribution intended for servers and workstations.

02 Fedora

fedora

Sponsored by Red Hat, is the foundation for the commercial RHEL

03 Debian

debian

Debian is an OS composed only of free, open-source software.

04 Others

Other popular distributions include:

Ubuntu, Linux Mint, CentOS, openSUSE / SUSE Linux Enterprise....

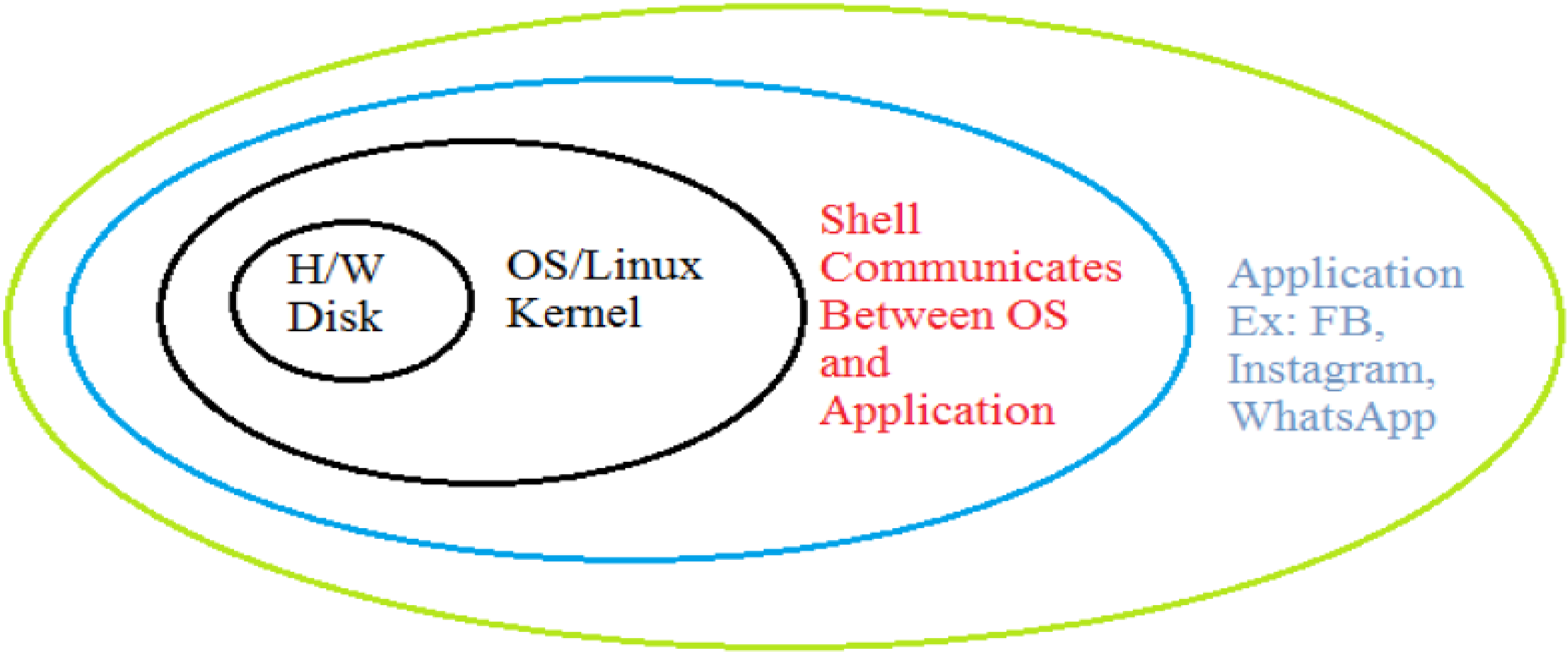
- Windows Server: Operating S/M

Graphical user interface

- Linux Server: Operating S/M

Command Line interface.

Linux Architecture



LINUX DIRECTORY STRUCTURE

Everything in Linux is a file or a directory. The entire file system starts from the root directory "/"

DIRECTORY TREE

/ Root Directory

- /bin
- /boot
- /dev
- /etc
- /home
- /lib
- /media
- /mnt
- /opt
- /proc
- /root
- /run
- /sbin
- /srv
- /sys
- /tmp
- /usr
- /var

Note: Some directories may not exist in minimal installations.

DIRECTORY	DESCRIPTION	EXAMPLE CONTENTS
/	The root directory. Top of the directory hierarchy.	bin, boot, etc, home, usr, var, ...
/bin	Essential user command binaries.	ls, cp, mv, cat, echo, ...
/boot	Boot loader files and kernel.	vmlinuz, initrd.img, grub/
/dev	Device files (hardware and virtual devices).	sda, null, tty, random, ...
/etc	System-wide configuration files.	passwd, fstab, ssh/, systemd/
/home	Home directories of regular users.	user1/, user2/, ...
/lib	Essential shared libraries and kernel modules.	libc.so, libm.so, modules/
/media	Mount point for removable media.	usb-drive/, cdrom/
/mnt	Temporary mount point for filesystems.	(usually empty)
/opt	Optional or third-party software packages.	google/, docker/, ...
/proc	Virtual filesystem with process and kernel info.	cpuinfo, meminfo, 1234/, ...
/root	Home directory of the root user.	.bashrc, .ssh/, ...
/run	Run-time system information since boot.	pid files, sockets, tmpfs data
/sbin	System binaries for system administration.	fdisk, reboot, ifconfig, ...
/srv	Data for services provided by the system.	www/, ftp/, ...
/sys	Virtual filesystem for kernel and devices info.	devices/, class/, ...
/tmp	Temporary files; cleared on reboot.	temporary files
/usr	User utilities, apps, libraries (multi-user programs).	bin/, lib/, share/, local/
/var	Variable data files (logs, caches, databases).	log/, cache/, lib/, spool/

Remember: The exact contents may vary between distributions, but this hierarchy is the standard.

Advantages of Linux :

1) Open Source (Free to Use):

- Linux is free and its source code is available
- Anyone can modify and distribute it

2) Highly Secure :

- Less affected by viruses and malware
- Strong user permission system
- Used in servers for safety

3) High Performance :

- Runs smoothly even on old or low-end computers
- Efficient use of system resources

4) Stable and Reliable :

- Rarely crashes
- Can run for long periods without restart

5) Customizable :

- You can change almost everything :
  - Desktop design
  - System behavior
  - Features

6) Multiuser and Multitasking :

- Multiple users can work at the same time
- Run many programs simultaneously

7) Supports Multiple Platforms :

- Works on:
  - Computers
  - Servers
  - Mobile devices (Android)
  - Embedded systems

Feature	Linux	Windows
Cost	Free and open-source	Paid (license required)
Security	Very secure, fewer viruses	More vulnerable to viruses
User Interface	Less beginner-friendly (varies by distro)	Easy and user-friendly
Performance	Fast, works on low-end systems	Requires higher resources
Customization	Highly customizable	Limited customization
Software Support	Fewer commercial apps	Wide range of software (MS Office, games)
Usage	Servers, developers, tech users	Personal use, business, gaming
Updates	Controlled by user	Automatic updates